

Pengfei Hu

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EDUCATION

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- Stevens Institute of Technology** Sep. 2023 – Present
Ph.D. in Computer Science Hoboken, NJ
- Advisor: [Dr. Yue Ning](#); GPA: 3.95 / 4.00;
 - **Research Interest:** Transfer Learning, Post-Training, Large Language Model, Graph Learning, Interpretability.
- University of Southern California** Sep. 2021 – May 2023
M.S. in Analytics Los Angeles, CA
- GPA: 4.00 / 4.00;
 - Research Assistant advised by [Dr. Sze-chuan Suen](#) working on monitoring of diabetes via wearable devices.
- Nanjing University of Finance and Economics** Sep. 2016 – Jun. 2020
B.S. in Economic Statistics Nanjing, China
- GPA: 3.63 / 4.00 (Rank 1st/150 in Major, National Scholarship).
 - Working on relative deprivation in the digital era ([one publication](#)) supervised by [Dr. Qinghai Li](#).
 - Interned at *Alibaba Group* (2021), *Tencent* (2020), *Didi Global* (2019–2020), and *PwC Consulting* (2019).

INTERNSHIP EXPERIENCE

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- Amazon Web Services** Sep. 2026 – Dec. 2026
Applied Scientist Intern New York City, NY
- **Mentor:** *Sébastien Jean*; **Manager:** *James Gung*;
 - **Topic:** Evaluating and Understanding memory system for proactive agents in long-horizon personal workflow.
- Oak Ridge National Laboratory** May 2025 – Aug. 2025
Research Scientist Intern Oak Ridge, TN
- **Mentor:** *Ming Fan*; **Leader:** *Dan Lu*;
 - **Project 1: Adaptive Graph Transformer for Multi-Reservoir Inflow Forecasting**
 - * Designed an adaptive graph learning module that captures latent inter-reservoir dependencies, coupled with a Transformer encoder to model spatiotemporal inflow dynamics in the Upper Colorado River Basin;
 - * **Paper: First-author** workshop paper at [IEEE International Conference on Data Mining \(ICDM\) 2025](#); extended version (**co-first-author**) published in [Environmental Modelling & Software](#) (IF: 4.6).
 - **Project 2: Hydrological Domain-Conditioned Modulation for Cross-Reservoir Generalization**
 - * Conditioned a shared backbone on reservoir meta-attributes (geography, storage, climate) via feature-wise linear modulation for zero-shot transfer to unseen reservoirs;
 - * **Paper: First-author** Oral at [AAAI 2026 Workshop on AI for Environmental Science](#).

SELECTED PROJECTS

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- Transparent Domain Adaptation via Concept-Grounded Orthogonal Inference**
[ICML 2026](#)
- Proposed an orthogonal and concept-grounded inference method for transparent adaptation in clinical data.
 - Designed a sparse autoencoder with a dictionary-induced metric to factorize patient representations into label-invariant and domain-covariant components via M -orthogonal residual projection.
 - Provided human-understandable explanations via sparse-dimension ablation while achieving superior performance over state-of-the-art DA baselines on two EHR datasets under spatial, temporal, and facility distribution shifts.
- LLM Agent for Causal Discovery in Diagnosis Prediction**
[EMNLP 2025 Findings](#)
- Designed an agent-based causal discovery framework leveraging large language models to explore disease relationships and improve interpretability and predictive accuracy in clinical diagnosis tasks.
 - Constructed a domain knowledge vector database using ChromaDB, integrating biomedical knowledge from Wikipedia and enhancing reasoning coverage via RAG over structured knowledge.
 - Implemented the agentic reasoning pipeline using LangChain and evaluated performance on public EHR datasets.

SKILLS

Foundation Models & Reasoning Techniques: LLM Post-Training (SFT, PPO, GRPO), Test-Time Reasoning, On-Policy Distillation, LLM Agents, Tool-Integrated Reasoning, Retrieval-Augmented Generation (RAG).

Infrastructure & Scalable ML Systems: PyTorch, Hugging Face Transformers, verl, verl-tool, TRL, Accelerate, SGLang, vLLM, PyTorch Geometric, TensorFlow, TensorBoard, Weights & Biases, AWS (EC2, S3).

Agent & Data Infrastructure: LangChain, ChromaDB.

Model Architectures: Transformers, Graph Neural Networks (GNNs), Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs), Sparse Auto Encoder.

Programming Languages: Python, R, SQL, Java, Latex.

SELECTED PUBLICATIONS

(The Google Scholar link can be found [here](#), * equal contribution)

First-authored Publications:

1. **Pengfei Hu**, et al., “Exploring Accurate and Transparent Domain Adaptation in Predictive Healthcare via Concept-Grounded Orthogonal Inference”, *International Conference on Machine Learning (ICML)*, 2026.
2. **Pengfei Hu**, et al., “Discovering Hierarchy-Grounded Domains with Adaptive Granularity for Clinical Domain Generalization”, *International Conference on Information and Knowledge Management (CIKM)*, 2026. **Oral**.
3. **Pengfei Hu**, et al., “Bridging Stepwise Lab-Informed Pretraining and Knowledge-Guided Learning for Diagnostic Reasoning”, *IEEE Journal of Biomedical and Health Informatics (IEEE JBHI, IF: 6.8)*, 2026.
4. **Pengfei Hu***, et al., “AGFormer: Adaptive Spatiotemporal Graph Informed Transformer for Multi-Reservoir Inflow Forecasting”, *Environmental Modelling & Software (IF: 4.6)*, 2026.
5. **Pengfei Hu***, et al., “MPLite: Multi-Aspect Pretraining for Mining Clinical Health Records”, *IEEE International Conference on Big Data (IEEE BigData 2024)*.

Collaborative Publications:

1. Xiaoxue Han, **Pengfei Hu**, et al., “Interpretable and Interactable Predictive Healthcare with Knowledge-Enhanced Agentic Causal Discovery”, *Findings of the Association for Computational Linguistics (EMNLP 2025 Findings)*.
2. Hsin-Ling Hsu, ..., **Pengfei Hu**, et al., “MEDPLAN: A Two-Stage RAG-Based System for Personalized Medical Plan Generation”, *Annual Meeting of the Association for Computational Linguistics (ACL 2025, Industry Track)*.

HONORS

- Best Doctoral Research Award at the iCNS @ Stevens AI Engineering and Science Symposium *Spring 2026*
- Stevens Institute of Technology Provost Doctoral Fellowship *2023 – 2028*
- University Outstanding Graduate (Summa Cum Laude) *Spring 2020*
- National Scholarship of Zhongli International (Top 0.5% in school) *Fall 2019*

PROFESSIONAL SERVICES

Teaching Experience:

- Teaching Assistant for CS-584: Natural Language Processing. *Fall 2024, Spring 2025*
- Teaching Assistant for ISE-535: Data Mining. *Spring 2022, Fall 2022*

Research Experience:

- Research Assistant in USC Viterbi School of Engineering. *Summer 2022*
- Research Assistant in Center for Economic and Social Research (USC Dornsife). *Fall 2022, Spring 2023*
- Graduate Researcher in Oak Ridge National Laboratory (GRO). *Summer 2025*
- Authorized User of the Frontier Supercomputer at the Oak Ridge Leadership Computing Facility.

Conference and Journal Review: NeurIPS 2026; ICML 2026; ACL Rolling Reviews (bimonthly since July 2025); KDD 2026, 2027; AAI 2026, 2027; MICAI 2025; CIBM; ACM TKDD; IEEE J-BHI; Journal of Hydrology; IEEE Access.